Experiment 5

Steps to Download Hadoop

Data Science Lab | Hadoop 3.4.1 on this Mac | Ojas Srivastava, U24AI028

1. Install Java

- brew install openjdk@17
- JAVA_HOME=/opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Contents/Home
- Check with: java -version

2. Download Hadoop

- Download the Hadoop 3.4.1 binary tarball:
- https://archive.apache.org/dist/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
- Extract it in the home folder:
- tar -xzf hadoop-3.4.1.tar.gz -C /Users/ojas
- Hadoop is then at /Users/ojas/hadoop-3.4.1
- Keep it outside the DS Lab folder. The space in that path breaks Hadoop.

3. Every time you open a terminal

- export JAVA_HOME=/opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Contents/Home
- export HADOOP_HOME=/Users/ojas/hadoop-3.4.1
- export PATH="$JAVA_HOME/bin:$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH"
- Check with: hadoop version

The version line should say Hadoop 3.4.1.

4. Start HDFS after the download

- core-site.xml: fs.defaultFS = hdfs://localhost:9000
- hdfs-site.xml: replication 1, NameNode and DataNode folders
- hdfs namenode -format once per fresh run (this wipes HDFS)
- hdfs --daemon start namenode
- hdfs --daemon start datanode
- Check with jps. It should list NameNode and DataNode.

5. Stop HDFS

- hdfs --daemon stop datanode
- hdfs --daemon stop namenode